

GROWTH

These are problems about growth functions of groups and related issues.

(G1) (S.I.Adian) Is it true that a finitely presented group has either polynomial or exponential growth? [Background](#)

(G2) (R.I.Grigorchuk) (a) Is it true that there is a gap between polynomial rate of growth and the rate of growth of the function $e^{\sqrt{n}}$ on the scale of growth of finitely generated groups?

(b) Is there a finitely generated group G whose growth function is equivalent to $e^{\sqrt{n}}$?

*(G3) (M.Gromov, J.Lafontaine, P.Pansu) Is there a group of exponential, but not uniformly exponential growth? [Background](#)

*(G4) (R.I.Grigorchuk) Is it true that every finitely generated infinite simple group has exponential growth? [Background](#)

(G5) (R.I.Grigorchuk) Is it true that every hereditary just infinite group (i.e., a residually finite group whose every subgroup of finite index is just infinite) has either polynomial or exponential growth?

(G6) (R.I.Grigorchuk) Is there a cancellative semigroup of subexponential growth whose quotient group (if it exists) has exponential growth?

(G7) (R.I.Grigorchuk) Let G be a group of subexponential growth, and let $f(n)$ be the number of different elements of length at most n in the group G . Does the limit (when $n \rightarrow \infty$) of the ratio $f(n+1)/f(n)$ always exist?

*(G8) Let $[F_r, F_r]$ be the commutator subgroup of the free group F_r of rank r , and let $f(n)$ be the growth function of $[F_r, F_r]$ in F_r (i.e., the number of different elements of length n in $[F_r, F_r]$). Find the asymptotics of $f(n)$. [Background](#)

(G9) Find the growth rate of the free metabelian group of rank r . [Background](#)

(G10) (R.I.Grigorchuk, I.Pak) Is it true that every group of intermediate growth contains two infinite subgroups A and B which commute elementwise, i.e., $ab=ba$ for any $a \in A, b \in B$? [Background](#)

(G11) (R.I.Grigorchuk, P. de la Harpe) (a) Calculate the growth series of Baumslag-Solitar groups $B(p,q)$ with $q > p > 1$.

(b) Part (a) is probably too hard in general; in fact, it is open even for $B(2,3)$. We single out this case as (allegedly) the most tractable one.

(c) At least, find the asymptotic rates of growth for $B(p,q)$. [Background](#)